Your name: _____, Roll no.: _____, Your phone no. (optional): _____

CSE 350/550: Network Security, END-semester exam: April 28, 2025, 90 minutes, 90 marks

Instructions:

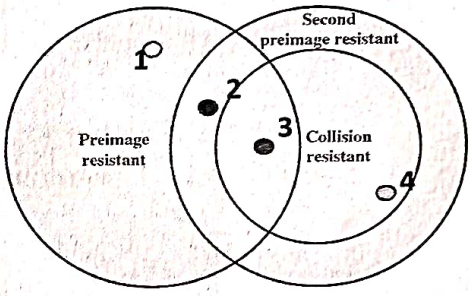
- The exam is in **two** parts: Part 1 consists of 6 questions that require short answers, each is for 5 marks → 30 marks. Part 2 consists of 5 questions that require relatively long answers, each is of 12 marks → 60 marks.
- This is a "closed-book" exam. All necessary information, including tables, etc. will be provided.
- Write your answers in the space provided. If necessary, use a separate sheet for rough work.
- Do not use unfair means. Action will be taken if you use unfair means. This includes using books, computer, phones, etc. or sharing information with other students. **You may use a calculator,** but nothing else.

Part 1: 6 questions that require short answers of 5 marks each

1. Hashing functions are classified based on whether the hash function satisfies one or more properties of "pre-image resistance", "second pre-image resistance" and/or "collision resistance" (see below). Clearly some hashing functions, such as '2', '3', and '4' given below, satisfy more than one property.
Question: where could the hash function(s), '1', '2', '3', or '4' be used to implement (a) one-way passwords, (b) detect file tampering, or (c) digital signatures? To answer this, fill the table below.

Hashing function		One-way passwords	File integrity	Digital Signature
	1	(Yes)/No	(Yes)/No	XXXXXXXX
	2	(Yes)/No	(Yes)/No	Yes/(No)
	3	(Yes)/No	(Yes)/No	(Yes)/No
	4	XXXXXXXX	(Yes)/No	(Yes)/No

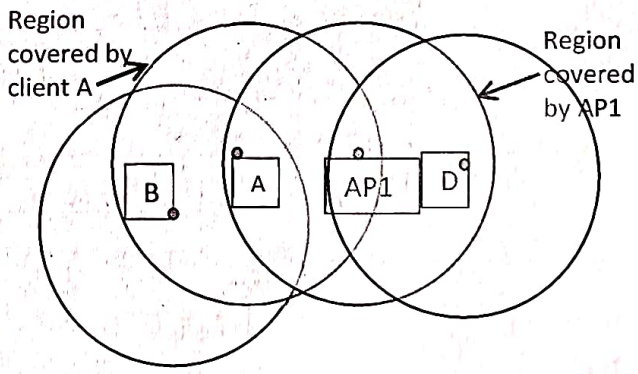
½ marks each
⇒ 5 marks



2. Consider the location of wireless devices, A, B and D, and access point, AP1 in a wireless network, shown below. Which set of devices(s) are visible or hidden from devices, A, B, D, or AP1? To answer, fill in the matrix with word 'visible' or 'hidden'? FYI, the circular region around a device at its center represents region covered by its signal.

½ marks each subject to max of 5 marks

Visibility	A	B	D	AP1
A		Visible	hidden	Visible
B	Visible		hidden	hidden
D	hidden	hidden		visible
AP1	visible	hidden	visible	



3. Assume that multiple messages, M1, M2, and M3, etc. are sent by Bob to Alice after encrypting them as C1, C2 and C3 using ElGamal cryptosystem and using the shared key, K, obtained using Diffie-Hellman cryptosystem. What is the perceived danger of using same shared key, K to encrypt and send M1, M2, and M3? Give details.

Clearly $C1 = KM1$, $C2 = KM2$, $C3 = KM3$, etc. If M1 is known at a later time, then an intruder can compute M2, M3, etc.

Since $\frac{C2}{C1} = \frac{KM2}{KM1} \Rightarrow M2 = C1^{-1} C2 M1$

Similarly $M3 = C1^{-1} C3 M1$, etc.

5 marks

4. Assume protocol WPA2 over TLS/SSL is used in IEEE 802.11 WiFi networks. **Question:** what kind of message should the client C send to an access point, AP, so that AP can authenticate C. See question 10 for a few exam messages.

3 marks

Client C → AccessPoint AP : $E(K_{PR-C}, [ID_C, AD_C, AP, N_1])$

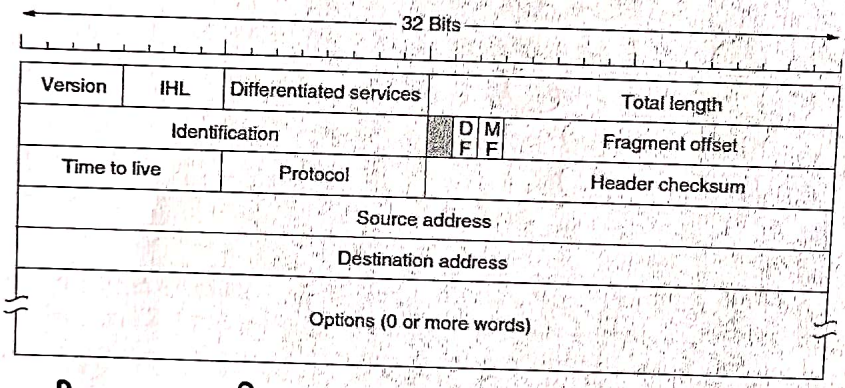
where K_{PR-C} = private key of client C, ID_C & AD_C are ID and IP address of C, N_1 is Nonce

AND what information should the access point AP possess so as to interpret the message from client C, and thereby authenticate client C?

2 marks

AP should possess the public-key of client C.

5. Assume IIITD establishes a 2nd campus in Noida, and connects the two LANs in Delhi & Noida by installing an IPSec "tunnel" between corresponding routers in Delhi & Noida. Before an IP packet exits the egress router in Delhi LAN, it is encrypted & encapsulated within a new IP packet. **Question**: Identify **4 fields** from the original IP header that are **NOT** copied into **IP header of outer IP packet? FYI, header of an IPv4 header is shown below.**



i Source IP address

ii Destination IP address

iii Time-to-Live

iv Header checksum

or even Total length

Identification

Any 4 of these will be good.

Award 1¼ marks each

⇒ 5 marks

6. Consider (as shown below) establishing IPSEC connections between the two campuses of IIITD **or** between a faculty workstation A and storage server V managed by academic administration. **Identify**:

   i. Security concerns that are addressed by the tunnel established between routers R1 & R2?

A. clients/servers outside the LANs, but connected to Internet will not know the original source/destination IP address of IP packets.

B. IP packets authenticated by ingress router R1, or R2, are decrypted and ~~inserted~~ allowed entry into the LAN.

4 marks

   ii. Which one or more devices (viz. A, B, C, D) are kept out of the tunnel between routers R1 & R2?

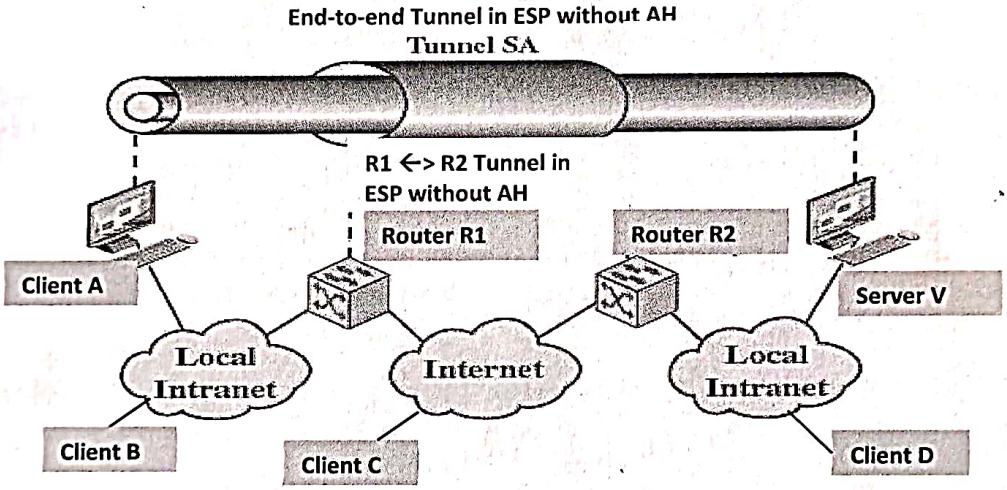
Client C

1/2 marks

   iii. Which one or more devices (viz. A, B, C, D) are kept out of the Transport connection between client A and server V?

Clients B, C, D

1/2 marks



Part 2: 5 questions that require somewhat longer answers (12 marks each)

7. Take a look at table below that describes a message together with the hash function/value. Here characters x_0, x_1, ... x_6 are message characters. The 8-bits of any character x_j are shown as: xj7, xj6, ..., xj0, **where xj7 is the (EVEN) parity bit obtained by adding bits row-wise as follows**. That is:

x07 + x06 + ... + x00 = 0; x17 + x16 + ... + x10 = 0; ..., x67 + x66 + ... + x60 = 0

A last character x_7 is added. It forms part of the hash value. It is obtained by adding bits column-wise as follows:

x70 + x60 + ... + x00 = 0; x71 + x61 + ... + x01 = 0; ...; x77 + x67 + ... + x07 = 0

Message characters	(parity) bit_7	bit_6	bit_5	bit_4	bit_3	bit_2	Bit_1	bit_0
x_0	x07	x06	x05	x04	x03	x02	x01	x00
x_1	x17	x16	x15	x14	x13	x12	x11	x10
x_2	x27	x26	x25	x24	x23	x22	x21	x20
x_3	x37	x36	x35	x34	x33	x32	x31	x30
x_4	x47	x46	x45	x44	x43	x42	x41	x40
x_5	x57	x56	x55	x54	x53	x52	x51	x50
x_6	x67	x66	x65	x64	x63	x62	x61	x60
(parity) x_7	x77	x76	x75	x74	x73	x72	x71	x70

To summarize, the sum of all bits in a given row is 0. And the sum of all bits in a given column is 0. The hash function consists of all the shaded bits from the table, viz. the bit_7 in each character, and the 8 bits in char_7. The rest is the original text.

I now prepare a message consisting of 7 characters, "I O U 1 2 0 0", where the characters in the message are encoded as 8-bits including the parity bit (as above). An 8 th character is added as above to complete the hash function/value. I now create an RSA-based digital signature using the above hashing function, H(x), with my private key PR-BNJ.

a. Give two examples where it is possible to replace "IOU1200" without this change being detected at the receiver's end? 4 marks

IOU2100, IOU1233

b. Give one example where a change will in fact be detected at the receiver's end? 2 marks

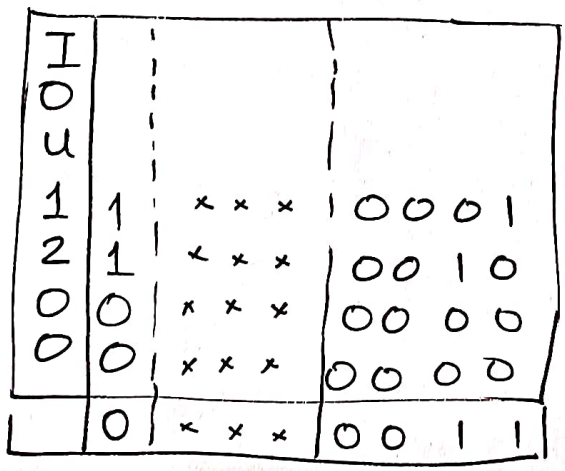
IOU2011

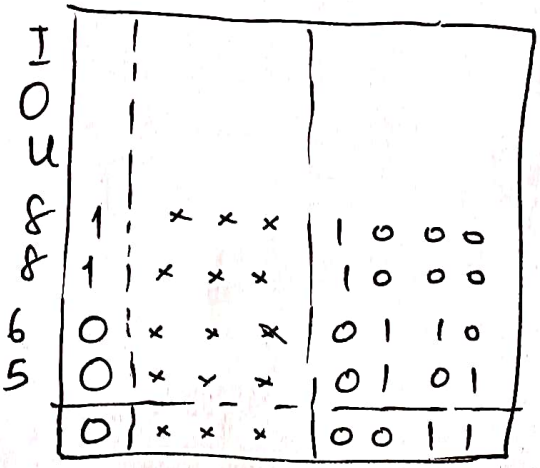
c. What change in the IOU note can be made by an intruder that will result in the largest value being owed by me (viz. the sender)? 6 marks

8865

Explain as to why this change will not be detected: The parity bits row-wise & column wise are the same

Concider IOU1200 & IOU8865





8. We propose that Bob use ElGamal Cryptosystem to send 2 messages, M1 & M2, to Alice. But, he uses Diffie-Hellman scheme to generate shared one-time shared keys, K1 and K2. The specific Diffie-Hellman scheme used assumes prime q = 11, and its primitive root a = 2.

a. Alice generates a private key XA = 8, computes and sends to Bob the corresponding public key YA.
b. As part of Bob trying to encrypt and send message M1 = 7,
   - he generates a private key XB1 = 3, and computes the corresponding public key YB1,
   - using Alice's public key YA, he computes the resulting shared key – in this case K1,
   - using the El-Gamal scheme, he encrypts message M1 = 7 using shared key K1, to obtain cipher C1, and
   - he sends the encrypted cipher C1 together with his public-key YB1.
c. Alice, on her part,
   - uses Bob's public key YB1 to compute the shared key (this better be = K1),
   - decrypts the encrypted cipher C1 received from Bob (again this better be = M1).
d. Steps b and c are repeated **when Bob wishes to send message M2 = 8, but using a different XB2 = 9.**

Show all your calculations below:

At Alice's end:
XA = 8; YA = 3;

$YA = a^{XA} = 2^8 = 3$

Calculations concerning **message M1 = 7**

At Bob's end:
XB1 = 3; YB1 = 8;
K1 = 5;
M1 = 7; C1 = 2;

$YB1 = a^{XB1} = 2^3 = 8$
$K1 = YA^{XB1} = 3^3 = 5$
$C1 = M1 * K1 = 7 * 5 = 2$

At Alice's end:
K1 = 5; inverse(K1) = 9;
Computed M1 = 7;

$K1 = YB1^{XA} = 8^8 = 5$
$K1 * K1^{-1} = 1 \Rightarrow K1^{-1} = 9$
$M1 = K1^{-1} * C1 = 9 * 2 = 7$

Calculations concerning **message M2 = 8**

At Bob's end:
XB2 = 9; YB2 = 6;
K2 = 4;
M2 = 8; C2 = 10;

$YB2 = a^{XB2} = 2^9 = 6$
$K2 = YA^{XB1} = 3^9 = 4$
$C2 = M2 * K2 = 8 * 4 = 10$

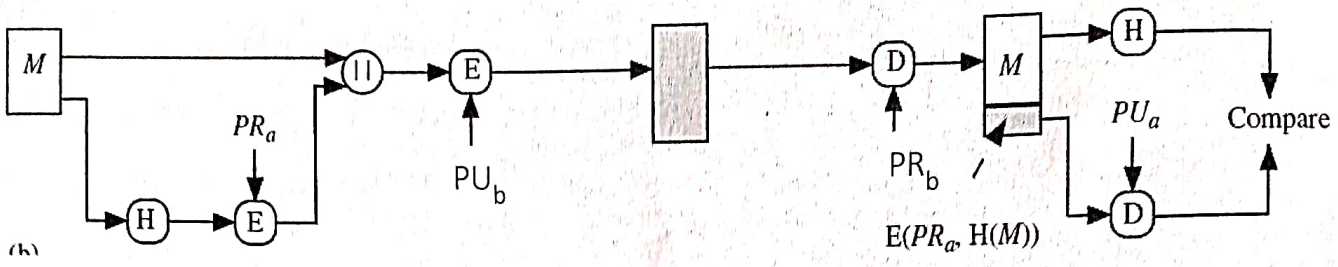
At Alice's end:
K2 = 4; inverse(K2) = 3;
Computed M2 = 8;

$K2 = YB2^{XA} = 6^8 = 4$
$K2 * K2^{-1} = 1 \Rightarrow K2^{-1} = 3$
$M2 = K2^{-1} * C2 = 3 * 10 = 8$

Correct answer 9 marks

detail 3 marks

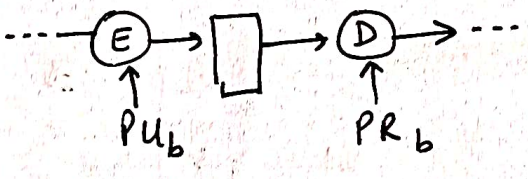
9. One way for Alice to send a message M to Bob is to (a) prepare message M, (b) compute its hash H(M), (c) sign H(M) using her private RSA key, PR-a, (d) encrypt the concatenated M || signed(H(M)) using Bob's public RSA key, PU-b, and (e) send the resultant to Bob. This is illustrated below.



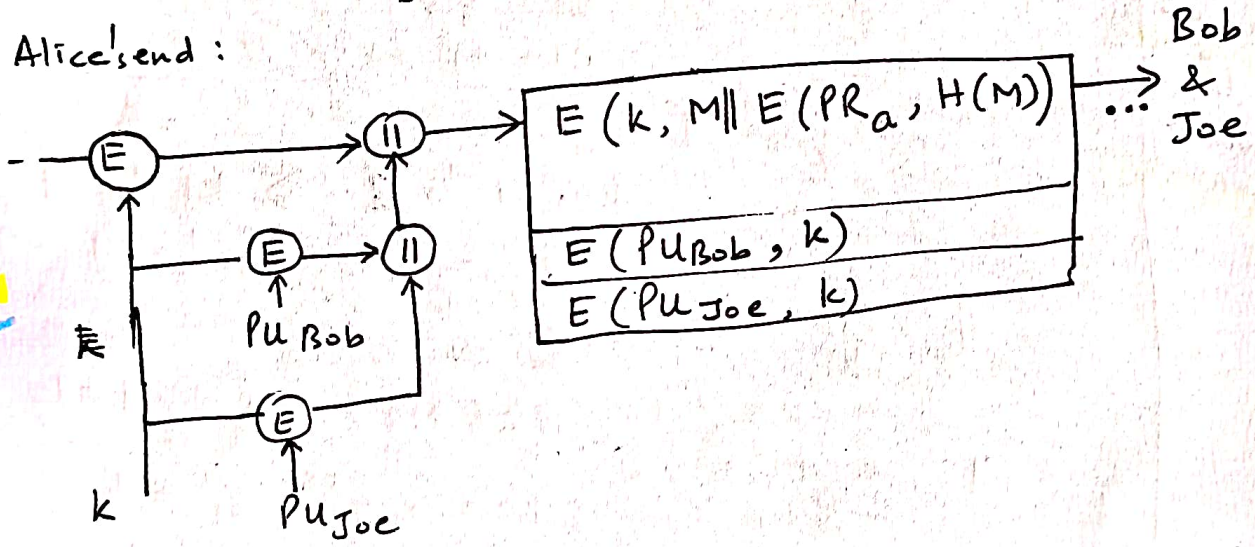
(b)

But instead, and presently, Alice wishes to send the message to two friends, Bob and Joe, simultaneously, without having to encrypt the concatenated M || signed(H(M)) twice. That being so, how can step (d) above be changed so that both Bob and Joe are able to decrypt the message? **Hint:** Alice encrypts the concatenated M || signed(H(M)) once using a session key, but copies that session key to both Bob and Joe confidentially. To answer the question, give a diagram suitably annotated.

As per the hint: the blocks are to be replaces as below:

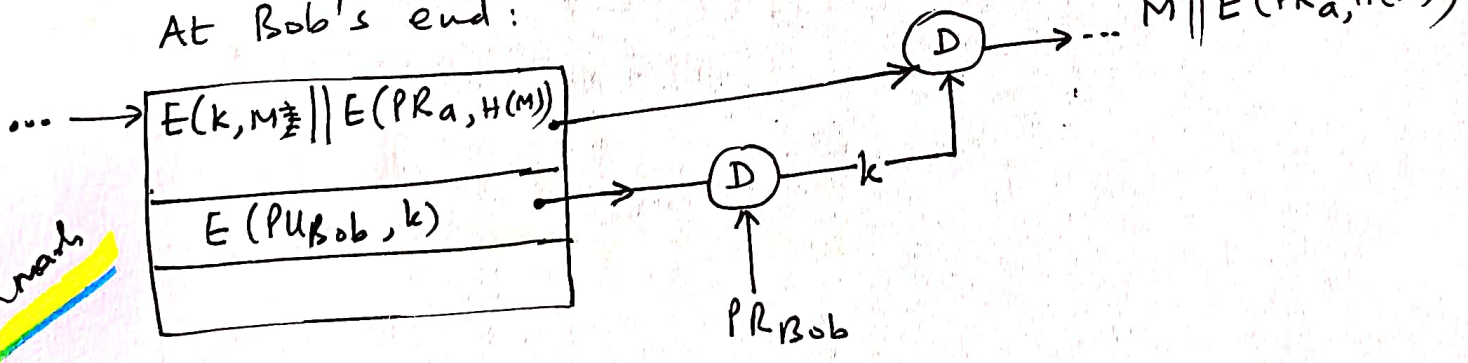


At Alice's end:



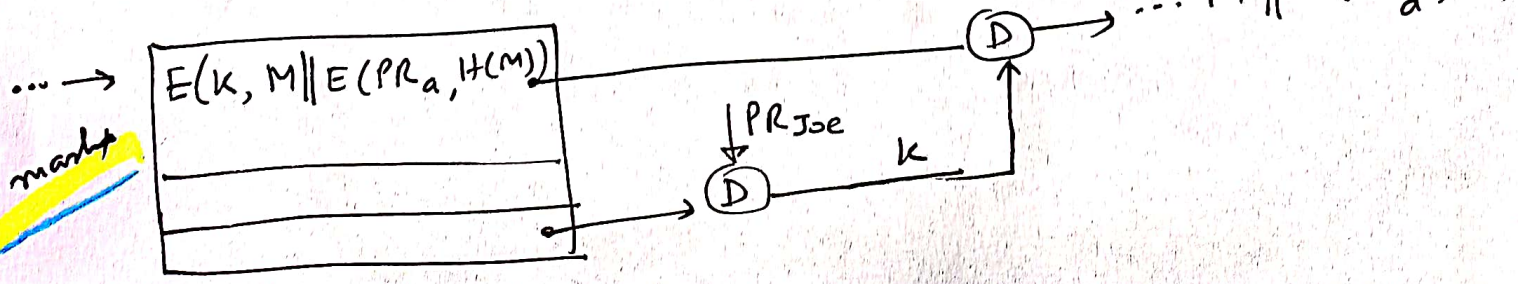
3 marks

At Bob's end:



2 marks

At Joe's end



2 marks

10. A portion of the Kerberos v4 protocol is given below. It is an incomplete description of the protocol in that the keys that are used by the client, the authentication server, or the ticket-granting server are simply specified as K_A, K_B, K_C, K_D, K_E. First figure out (and then briefly explain) **what purpose these keys serve**, and whether (and how) **keys K_A, K_B, K_C, K_D, K_E are related** to each other. Note, the timestamps, etc. have been omitted. And interaction related to server V are also omitted.

1. C → AS — $ID_C \,||\, ID_{TGS}$
2. AS → C — $E(K_A, [K_B \,||\, ID_{TGS}].\ Ticket_{TGS})$
   where $Ticket_{TGS} = E(K_C, [K_D \,||\, ID_C \,||\, AD_C \,||\, ID_{TGS}])$
3. C → TGS — $ID_C \,||\, Ticket_{TGS} \,||\, Authenticator_C$
   where $Authenticator_C = E(K_E, [ID_C \,||\, AD_C])$
4. TGS → C — etc.

10 marks

K_A = Kc the key shared between C & AS

K_B = Kc,tgs the key to be used by C to help 'tgs' authenticate itself and authenticate 'tgs'

K_C = ~~Ktgs the shared key between~~ AS and tgs.

K_D = ~~Kc,tgs the key used by C &~~ tgs to authenticate each other

K_E = ~~Kc,tgs the key as used~~ above.

Relations (if any) between K_A, K_B, K_C, K_D, K_E

2 marks

~~$K_B = K_D = K_E$~~

11. An email addressed to user@cs.vu.nl is to be transferred from the sending mail server to the receiving mail server after undertaking several steps. Some of these are mentioned below. NOTE: these steps are NOT in the proper sequence. ALSO, some information is missing for each step. Other than completing the missing information for each step, state the sequence in which the 8 steps are undertaken. Supply your response in the table below.

Incorrect sequence	Step to be undertaken	Fill in the missing information here using expressions such as "list of headers", "MX record", "email address", "DNS server", "ADMD", etc.	Correct sequence
1	Determine the IP address of the mail server using:	DNS server for mailserver for ADMD = cs.vu.nl	4
2	Access the DNS server relevant to:	ADMD = cs.vu.nl	2
3	Determine the URL of mail server from:	MX record for ADMD from DNS for ADMD	3
4	Transfer to destination ADMD original headers and body together with:	DKIM-signature of originating ADMD, including hash value & signature	8
5	Compute the signature by encrypting the:	hashvalue calculated earlier	7
6	Compute the destination administrative mail domain (ADMD) from:	email address, viz. user@cs.vu.nl.	1
7	Compute the hash value from the given: identified	headers and body of email	5
8	Identify the key to be used to sign the:	DNS server for domain ADMD = cs.vu.nl	6

1½ marks each — ½ for correct sequence; 1 for correct description

= 12 marks